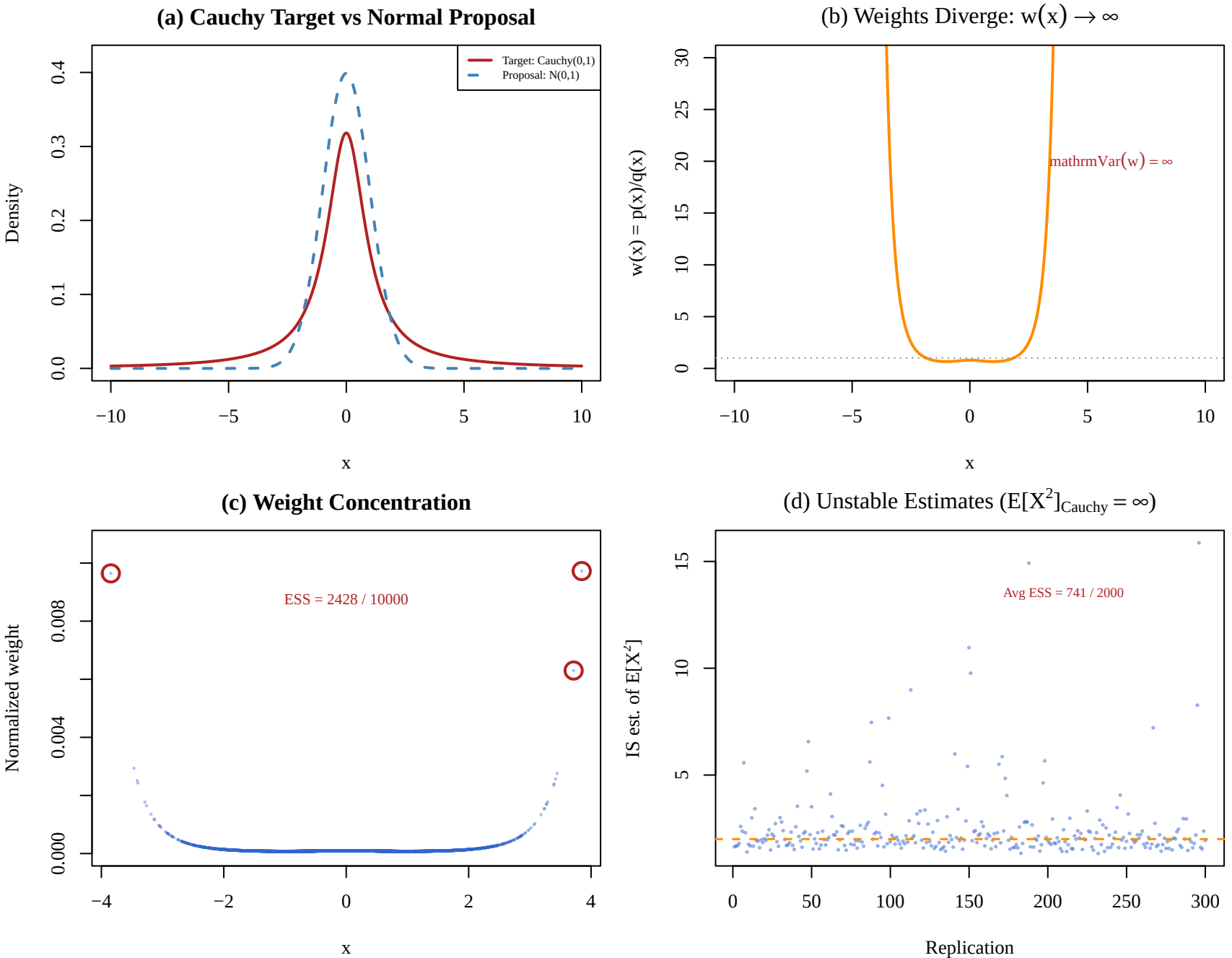
(a) Cauchy Target vs Normal Proposal
Target: Cauchy(0,1)
Proposal: N(0,1)
Density
x
(b) Weights Diverge: w(x) → ∞
mathrmVar(w) = ∞
w(x) = p(x)/q(x)
x
(c) Weight Concentration
ESS = 2428 / 10000
Normalized weight
x
(d) Unstable Estimates (E[X²]Cauchy = ∞)
Avg ESS = 741 / 2000
IS est. of E[X²]
Replication